

NIMBUS-9¾ BROOM

Manned eVTOL — 1 rider, 8 ducted fans, fly-by-intent

Project NIMBUS-9¾ · Real Quidditch · hardware full-stack

ISOMETRIC

SIDE

TOP

CUTAWAY — internal drive revealed

KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Overall length2400 mm

core/params.py LEN = 2.4 m

Lift fans8 × EDF (internal)

_default_fan_layout (exact x/y)

Thrust-to-weight2.45 (hover @ 41%)

analysis/flight_thermal.py

All-up mass120 kg

rider ~80 + airframe ~40

Battery sprint~61 s / charge

2600 Wh alone — a sprinter

Match endurance~26 min (hybrid)

series-hybrid: analysis/endurance_match.py

Cooling margin8.1× waste heat

lift air = coolant + heat pipes

Control inputwave-to-fly (gesture)

firmware/rider_input.py -> intent mapper

Control coreCortex-M4F @ 400 Hz

rust/nimbus_core + firmware/hal.py

NOTE

Disc area is capped by what fits inside the bristle flare (the fan never protrudes), so battery-only it's a ~60 s sprinter. A slim in-shaft SERIES-HYBRID genset burning SAF lifts that to a ~26 min match (the battery becomes the peak buffer; full ~250 kW bursts kept). Fans internal, lift air = coolant. And you fly it by WAVING the handle pose -> RiderIntent -> the existing intent mapper. FLIGHT + THERMAL + MATCH all PASS.

BILL OF MATERIALS — BRM (38 line items, prototype qty 1)

REF	PART	SPEC	QTY	\$ EA	\$ EXT
BRM-01	Carbon-fibre monocoque body p...	2.4 m, hides the fan bay; SI->mm x1000 in cad/broom.scad	1	2,200	2,200
BRM-02	Handle shaft + brass pommel	tapered Ø34->Ø22 mm, leather-wrapped grip	1	180	180
BRM-03	Bristle fairing leaf	~780 carbon-straw strands, aerodynamic shroud	1	640	640
BRM-04	Electric ducted fan (EDF) unit	360 N max each; main-lift in bristle shroud + in-shaft trim fans	8	520	4,160
BRM-05	Brushless motor for EDF	~12 kW peak, 6S-14S, duct-matched	8	340	2,720
BRM-06	ESC (electronic speed control...	200 A, telemetry, 1st-order ~40 ms response	8	150	1,200
BRM-07	Li-ion buffer battery pack	2600 Wh, serves the ~250 kW peaks in the hybrid	1	3,100	3,100
BRM-08	Battery management system (BM...	14S smart BMS, cell balancing + SoC telemetry	1	260	260
BRM-09	Power distribution board	800 A bus, redundant rails	1	180	180
BRM-10	Flight controller MCU	Cortex-M4F (thumbv7em-none-eabihf), 400 Hz loop	1	95	95
BRM-11	IMU (redundant)	triple-redundant 6-DOF, 1 kHz	3	55	165
BRM-12	Magnetometer	3-axis, maneuver-compensated	2	18	36
BRM-13	Barometer	high-res, for altitude hold	2	12	24
BRM-14	RTK-GPS receiver + antenna	cm-grade, dual-band	1	420	420
BRM-15	UWB ranging tag	indoor cm positioning to pitch anchors	2	40	80
BRM-16	Solid-state LiDAR	360deg, edge collision avoidance	2	600	1,200
BRM-17	Low-latency datalink radio	redundant 5G + dedicated mesh	2	140	280
BRM-18	Ballistic recovery parachute ...	low-altitude rated, pyro deploy	1	1,500	1,500
BRM-19	Airbag skirt	underbody impact airbag, impact-triggered	1	450	450
BRM-20	Master kill-switch receiver	referee e-stop -> gentle descent	1	60	60
BRM-21	5-point harness + saddle	straddle seat, energy-absorbing	1	320	320
BRM-22	Rider intent controls	handlebar grips, throttle, lean sensors	1	210	210
BRM-23	Brass compass/attitude gauge	cosmetic + real attitude readout	1	90	90
BRM-24	LED trajectory strip	addressable, night-match light show	1	70	70
BRM-25	Heat pipe (sintered copper)	Ø6 mm, full-shaft length, motor->shaft spreader	4	22	88
BRM-26	Air-over-motor duct shroud	routes lift airflow across windings (air = coolant)	8	16	128
BRM-27	In-duct winding temp sensor	NTC, one per fan, feeds the governor	8	3	24
BRM-28	Thermal interface material	phase-change pad, motor + ESC to spreader	1	18	18
BRM-29	DShot ESC wiring harness	bidirectional DShot600, 8 channels + telem	1	45	45
BRM-30	CAN-FD vehicle bus	5 Mbit, BMS + ESC telem + pose to Dementor	1	30	30
BRM-31	Hardware abstraction layer	firmware skin around the Rust core	1	0	0
BRM-32	Micro-turbine genset	~125 kW continuous, slim in-shaft series-hybrid range extender	1	9,800	9,800
BRM-33	FADEC + start system	full-authority turbine control + spin-up	1	1,200	1,200
BRM-34	SAF fuel bladder + lines	~18 L crash-safe bladder inside the shaft	1	340	340
BRM-35	Sustainable aviation fuel	15 kg SAF per fill (~48x li-ion energy density)	1	30	30
BRM-36	Exhaust ejector + heat exchan...	entrains bypass air, drops exhaust <150 C	1	420	420
BRM-37	Ceramic-matrix bristle tips	high-temp bristles at the exhaust path	1	260	260
BRM-38	Gesture input transducer	handle IMU + twist grip -> RiderIntent	1	0	0

UNIT TOTAL (prototype):

\$32,023